

SAHIL NEGI

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Education

Year	Degree/Exam	Institute	CGPA/Marks
2022–Present	B.Tech in CSE	Graphic Era Hill University, Dehradun	7.69
2021	C.B.S.E	Holy Angel Sen. Sec. School	8.1
2019	C.B.S.E	Holy Angel Sen. Sec. School	9.01

Projects

Virtual Assistant

- Built an Agentic Graph RAG system integrating FAISS and Neo4j to enable context-aware and relationship-driven responses beyond traditional RAG limitations.
- Used an LLM-based agent using LangChain that dynamically selects tools (RAG and graph queries) to perform multi-step reasoning and improve decision-making accuracy.
- Designed a hybrid retrieval pipeline combining semantic similarity search with graph traversal to capture entity relationships and enhance answer relevance.
- Implemented a FastAPI backend for real-time query processing to provide scalable and low-latency AI responses.

Auto Scaling Engine

- Developed a full-stack auto-scaling system using React and FastAPI to monitor and manage dynamic workloads
- Built a rule-based prediction system to simulate intelligent scaling decisions based on CPU, memory, and request load
- Improved system reliability by addressing CORS issues, API failures, and UI crash scenarios
- Implemented a dynamic worker pool using a producer-consumer model, where scaling decisions directly controlled the number of active processing threads.

Stock Market Analysis & Prediction

- Built a stock prediction model using Time Series Analysis and Models like (LSTM, ARIMA), trained on 5+ years of market data.
- Integrated technical indicators (RSI, MACD, moving averages) with Future Price Prediction, achieving 85% accuracy.
- Enabled trend forecasting and decision support for short- and long-term investments.

PublicEye

- Developed a full-stack web application using React.js (frontend) and FastAPI (backend) for real-time incident reporting and visualization.
- Designed and integrated RESTful APIs for handling user data, reports, and geolocation-based services.
- Integrated MongoDB using async drivers (Motor) for efficient, scalable data handling and using modular architecture (routes, services, models, utils) for maintainability.

Technical Skills

Languages: Python, Java, C++, JavaScript, HTML , CSS

AI/ML/DL: Pandas, Numpy, Matplotlib, Tensorflow

Frameworks/Tools: FastAPI, React, Node.js, LangChain

Databases: MySQL, MongoDB

Activities

- Active member of the Debating Society, enhancing public speaking, argumentation, and critical thinking skills through inter-college competitions
- Participated in hackathons, collaborating in fast-paced environments to design and deliver innovative technical solutions under strict deadlines .
- Contributed as a volunteer in NGO initiatives, supporting community development through awareness campaigns and social impact activities.
- Represented teams in football tournaments, demonstrating leadership, teamwork, and strategic decision-making abilities.